



TEAM OFFGRID / DEFENSE &  
DISASTER ROBOTICS

AUTONOMOUS  
SWARM

AUG 2026  
• REV  
1.0

*“When GPS fails and RF links jam,  
SUTRA geonavigates and locates  
survivors in real-time.”*

# PROJECT SUTRA.

Swarm Unified Tactical Reconnaissance Architecture  
— decentralized multi-UAV flight, Deep JSCC neural  
zero-cliff video, and 3.59cm terrain-corrected DEM  
survivor geolocation.

## DECENTRALIZED GNC

**50Hz** OFFBOARD  
RATE

PX4 MicroXRCE + VIO  
EKF2 Odometry

## SEMANTIC WIRELESS

**-5dB** JAM  
RESILIENCE

Deep JSCC Neural  
Zero-Cliff Video

## TERRAIN GEOLOCATION

**3.59cm** GEO-  
ACCU

WGS84 3D DEM  
Terrain Raycasting

CORE ARCHITECTURE TEAM  
(OFFGRID)

RESEARCH-BACKED & EMPIRICALLY  
VALIDATED

### Nikhil

Tech Lead ·  
Subsys A &  
B

GNC &  
JSCC

### Vedanth Sai Ram

Lead ·  
Subsystem  
C

### Siva Kesava

Lead ·  
Subsystem  
D

### Harika

Lead ·  
Subsystem  
E

VERIFICATION  
QA

### Rohith Kumar

Lead ·  
Subsystem  
F

AI  
PERCEPTION

3D GIS  
GCS

NDMA  
CONOPS